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Advanced Glycation End Product (AGE): Characterization of the Products From the Reaction Between D-Glucose and Serum Albumin

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We incubated bovine serum albumin (BSA) with glucose in an attempt to study how the advanced glycation end products (AGEs) are formed and what methods can be used for their identification and isolation. The reaction was monitored by boronated affinity gel, size exclusion and ion exchange chromatography, and chromatofocusing. Reaction products were also characterized by fluorescence measurement, fructosamine assay, and polyacrylamide gel electrophoresis (PAGE). Based on the measurement of AGE-associated fluorescence (excitation, 370 nm; emission, 440 nm) we found that the AGEs could be detected as early as after 3 days incubation. The fluorescence was always associated with the larger molecules of cross-linking product resulting from the reaction between BSA and glucose. The overall fluorescence intensity increased with incubation time and fluorescence of the highest intensity was found with the AGE product largest in size. As with the Amadori product, AGEs also bind to the boronated gel column but with an even higher affinity. Compared to the original albumin monomer AGE molecules are not only larger in size but also have lower isoelectric points and carry more negative charges. Both the size and the negative charges of AGEs continue to increase over time during incubation. This results in a group of cross-linking molecules heterogeneous in size and charge. These results will aid in both the isolation and selection of appropriate AGE molecules for the preparation of anti-AGE antibodies, calibrator, and control in the development of an AGE immunoassay.

Key words: diabetes, fluorescence spectrum, chromatography, PAGE, heterogeneity

INTRODUCTION

Proteins containing free amino groups, such as the ε-NH_2 groups of lysine and arginine at physiological pH will react non-enzymatically with sugar aldehydes or ketone groups, such as glucose, to form a reversible Schiff base, which rearranges in a few hours to a stable, covalently bonded Amadori product, a protein-sugar adduct (1–3). Amadori products of hemoglobin and albumin are routinely measured in clinical laboratories for determining glycemic control in patients with diabetes. What may not be generally understood is the fact that the Amadori products will undergo further rearrangement to produce a heterogeneous group of polymers called advanced glycation end-products or AGEs. AGEs exhibit various physicochemical properties which are quite different from the parent proteins. The most obvious changes observed in AGEs are derived from the ability of AGE to cross-link with other proteins. As a consequence, the molecular sizes of AGEs are much larger than the parent proteins and are heterogeneous in size and composition. Formation of AGE can also be detected by its characteristic fluorescence spectrum (excitation, 370 nm; emission, 440 nm) and its yellow-brown color (1–4).

It appears that there is a common epitope expressed among all AGEs. Antibodies prepared against any of these AGEs will react with other AGEs regardless of their parent proteins (5, 6). For example, antibodies prepared against AGE-bovine serum albumin (AGE-BSA) not only react with AGE-BSA, AGE-human serum albumin, and AGE-hemoglobin but also with α-tosyl-1-lysine, α-tosyl-1-lysine methyl ester, monoaminocarboxylic acid, γ-amino-n-butyric acid, and β-alanine (7). These antibodies will also recognize the products if some amino acids and monoaminocarboxylic acids are used to substitute for the protein in the AGE molecule or if glucose is replaced with either fructose, glucose 6 phosphate aldoses and aldehydes. Anti-AGE antibodies, however, do not cross-react with Amadori products.

Abbreviations: AGE, advanced glycation end product; BSA, bovine serum albumin; PAGE, polyacrylamide gel electrophoresis; SDS, sodium dodecyl sulfate.

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AGEs have been shown to be present in the plasma (8) and to accumulate in the tissues (9). Many studies suggest that AGEs play an important role in the structural and functional alterations of proteins during aging and long-term diabetes. When proteins with long half-lives, such as collagen, crytallins, elastin, and myelin, are involved with AGE formation, they will continue to cross-link to other proteins and lead to many aging and diabetes associated dysfunctions and complications (1–3, 9, 10). Persistent accumulation of circulating proteins by AGE-collagen, such as serum albumin, may contribute to the characteristic thickening of diabetic basement membranes. AGEs are also recognized and removed by scavenger receptors on endothelial cells and monocytes (6, 11–13). For example, removing glycated low density lipoprotein (LDL) by macrophages leads to the pathogenesis of atherosclerosis.

AGE could also contribute to Alzheimer's disease (AD). Recently a higher concentration of AGE adducts was actually found in plaque fractions of Alzheimer's disease (AD) brains. Aggregation of soluble β-amyloid peptide (βAP) in vitro was found to be enhanced by the addition of small amounts of pre-aggregated βAP "seed" as well as by the AGE-modified βAP-nucleation seeds (14, 15). The stability of plaques and tangles and their high content of lysine in AD make them ideal candidates for glycation. Aggregation of soluble β-amyloid (Aβ) was also recently found to be promoted by AGE-modified synthetic Aβ. The importance of AGEs to AD is also supported by the finding of apo E4 in senile plaques and neurofibrillary tangles of AD brains. The apo ε allele is a risk factor in both sporadic and familial late-onset AD. The apo E4 isoform may promote the formation of AGEs and then the plaque in AD brains because Aβ binds selectively to the apo E4 isoform and the apo E is highly susceptible to glycation due to its higher arginine content.

Because the formation of AGEs from the Amadori product can proceed in the absence of glucose, an improvement of glycemic control alone may not be sufficient to prevent the continued progression of this pathologic process. Conceivably, administration of pharmacological agents, such as aminoguanidine, that will directly interfere with the formation of AGEs is the optimal future therapy for the prevention of hyperglycemia-initiated tissue damage and clinical complications (16). An assay for AGE could be used to identify elderly patients and patients with diabetes who may be at risk for developing AGE-associated clinical complications. The AGE assay could also be used to monitor the success of therapy and is a useful tool for identifying medications better than aminoguanidine. The anti-AGE antibodies could also be applied to immunohistochemical procedures in tissue specimens.

In the present study our objective was to identify and isolate AGEs for the preparation of anti-AGE antibodies, standards, calibrators, and controls to be used in the development of an immunoassay for AGE. In the past, several investigators have produced AGEs by incubating albumin with glucose (5, 7). Usually the entire incubation mixture of albumin and glucose, after dialysis, was injected into the animals for preparation of anti-AGE antibodies or coated on the microwell for the establishment of an immunoassay. We believe a purified fraction of AGE from the incubation mixture should lead to more specific antibodies. Therefore, in the present investigation, we incubated BSA with glucose and followed the reaction with affinity chromatography and a fructosamine assay and monitored AGE-associated fluorescence for the formation of Amadori product and eventually the AGEs. All products in the incubation mixture at various time intervals were separated and characterized by varying chromatographies and were identified by PAGE.

MATERIALS AND METHODS

DEAE Sephacel gel, Sepharose CL-4B, Pharmacia PBE 94 gel, polybuffer 74, and Superose 12 HP 10/30 prepacked columns were all purchased from Pharmacia (Alameda, CA). The BCA protein assay reagent (bicinchoninic acid) was from Pierce Chemical Co. (Rockford, IL). All chemicals and buffers used for the PAGE came from Bio-Rad Laboratories (Hercules, CA). The m-aminophenyl-boronated-agarose gel (Glycogel) and the Pierce GlycoTest™ II kit for affinity chromatography were obtained from Pierce (Rockford, IL). Both BSA (bovine serum albumin) and D-glucose were purchased from Sigma (Sigma, St. Louis, MO). The Glyco-PROBE™ GSP kit used for the fructosamine assay was from ISOLAB Inc. (Akron, Ohio).

Incubation of BSA With D-Glucose

The incubation mixture contained 900 mg D-glucose, 500 mg BSA, 0.02% (W/V) sodium azide, and 0.01% (W/V) gentamicin sulfate in 0.2 M sodium phosphate buffer, pH 7.4 (final concentrations: 0.5 M glucose and 50 mg/mL BSA); this is similar to the procedure described by Makita et al. (5). The final volume was 10 mL. The buffer was sterilized by ultrafiltration through a 0.2 μ filter (Nalge, Rochester, NY). The bottle containing the mixture was sealed and incubated at 37°C in the dark. Aliquots were withdrawn at various time intervals and stored at –70°C then analyzed at the same time. A control, which contained all the ingredients except the glucose, was also incubated under the same condition at the same time.

Fluorescence Measurement

The fluorescence of the incubation mixture and column eluates was measured on a FLUORO IV™ fluorescence spectrometer (GILFORD, Oberlin, OH). Except when scanning was performed, the fluorescence was measured at 370 nm for excitation and 440 nm for emission wavelength. The protein concentration of each specimen was always adjusted to 1 mg/mL before use for fluorescence determinations. To standardize the instrument, a freshly prepared BSA solution (1 mg/

mL) was used to adjust the meter to zero fluorescence and the mixture, after 60 days incubation, was used to calibrate the meter to 100 fluorescence units.

Boronated Gel Affinity Chromatography

The amount of glycated BSA formed during incubation at various time intervals was determined by boronated gel affinity chromatography following the kit instruction. A column packed in-house with 5 mL of m-aminophenylboronated agarose gel was also used in order to obtain sufficient quantity of glycated proteins for fluorescence measurements. A buffer containing 0.25 M ammonium acetate, 0.05 M magnesium chloride and 0.5% phenoxyethanol, pH 8.0 was used for column equilibration and for eluting nonglycated proteins. The bound proteins were eluted with a 0.1 M Tris-HCL buffer containing 0.2 M sorbitol and 0.5% phenoxyethanol, pH 8.5. After the majority of bound proteins were eluted, 0.5% acetic acid was then used to regenerate the column and also served to elute any tightly bound proteins.

Fructosamine Assay

Fructosamine (1-amino-1-deoxy-fructose) was determined by a non-kinetic colorimetric method using the Glyco-PROBE™ GSP kit on a microplate following kit instructions (17); 50 μL of 10-fold diluted incubation mixture was used for each determination. Dilution was made by 0.2 mol/L phosphate buffer, pH 7.4. This reaction takes advantage of the reducing ability of the Amadori product at alkaline conditions, where it reacts with only glycated albumin and not the AGEs.

Polyacrylamide Gel Electrophoresis (PAGE)

Electrophoresis was carried out on the polyacrylamide gel plate using the Hoefer SE 250 Mighty Small II vertical slab unit (San Francisco, CA) following essentially the procedure described by Laemmli (18). Approximately 20 to 25 mg of total protein was applied to each trough on top of the gel plate. The tank buffer contained 25 mM Tris and 192 mM glycine, pH 8.3. The electrophoresis was carried out at constant voltage (120 V) at 8°C. The gel was stained overnight with one-step Coomassie Blue stain [0.003% Coomassie Brilliant Blue R-250 in methanol/water/glacial acetic acid (45/45/10. v/v/v)]. The gel was then preserved with preserving solution [10% glycerol in methanol/water/glacial acetic acid (45/45/10, v/v/v)] for 30 min and air dried.

SDS-PAGE

SDS-PAGE was also performed in the SE 250 Mighty Small II Vertical Slab Unit following the procedure described by Weber and Osborn (19). The samples, containing approximately 20–25 μg of protein, after mixing with an equal volume of 2-fold diluted sample buffer (125 mM Tris-Cl, 4% SDS, 20% glycerol, 10% dithiothreitol, and 0.01% bromophenol blue, pH 6.8), were boiled for 5 minutes and then applied to individual troughs of the gel. The tank buffer consisted of 25 mM Tris, 192 mM glycine, and 0.1% SDS, pH 8.3. Electrophoresis was carried out at constant voltage (120 V) and maintained at 8°C. At the end of electrophoresis the gel was stained and preserved following the same procedure described above.

Size Exclusion-HPLC

The change in size of the BSA molecule during incubation with glucose was monitored by HPLC (Ranin Instrument) on a prepacked column (10/30) containing Superose 6 HR from Pharmacia. A buffer containing 0.05 M phosphate and 0.15 M sodium chloride at pH 6.8 was used for column equilibration and elution. Prior to column injection, any particles present in the sample were removed by centrifugation. The flow rate was controlled at 0.5 mL/min and the eluates were collected at 1 mL/2 min/tube. The protein concentration of eluate was monitored using the on-line UV/vis detector at 280 nm.

Anion-Exchange-HPLC

The anion-exchange chromatography was also performed on a Ranin HPLC system using a Hydropore AX 83-603-ETI prepacked column from Rainin (Woburn, MA). The buffer containing 0.01 M phosphate buffer at pH 7.1 was used for column equilibration and sample dialysis. The same buffer containing additional 1 M NaCl was used for elution. All samples were dialyzed against the equilibration buffer and centrifuged before injection. For each run approximately 100 to 180 mL of sample was injected. After washing with equilibration buffer for 42 minutes the column was further eluted with buffer containing 0.01 M phosphate and 1 M sodium chloride, pH 7.1. The flow rate was controlled at 0.5 mL/min and the eluate was collected at 1 mL/2 min/tube. The protein concentration was monitored using the on-line UV/vis detector at 280 nm or by the BCA protein assay. The conductivity of the eluates was measured using a CDM 3 conductivity meter (Radiometer Copenhagen, Denmark).

DEAE Sephacel Chromatography

DEAE Sephacel chromatography was carried out at 4°C on a Pharmacia column (K 9/15 or 0.9 × 15 cm) containing about 9 mL DEAE Sephacel from Pharmacia (20). A linear gradient of NaCl from 0 to 0.5 M in the equilibrium buffer (0.01 M phosphate, pH 7.2 at 4°C) was used for protein elution. Eluates were collected at approximately 3.3 mL/100 drops/tube.

Chromatofocusing

A Pharmacia FPLC column (5/5 HR, 0.92 mL bed volume) was packed in-house with Pharmacia PBE 94 gel and used for chromatofocusing on a Rainin HPLC system (Emeryville, CA) over the pH range 7–4. For column equilibration and sample dialysis, a buffer containing 0.025 M imidazole buffer (pH 7.1) was used. All specimens, after dialysis, were filtered through 0.2 μm centrifuge filter tube to remove particles before HPLC injection. Polybuffer 94-HCl was used for elution after being diluted with water (8-fold) and adjusted to pH 4. At the end of elution an additional 1 M NaCl aqueous solution was used to wash out any protein remaining in the column. The flow rate was maintained at 0.5 mL/min and the eluates were collected at 1 mL/2 min/tube. We followed essentially the instructions for chromatofocusing provided by Pharmacia LKB Biotechnology.

Sepharose CL-4B Chromatography

Changes of the size of the BSA molecule incubated with D-glucose for 3 and 50 days were also examined by Sepharose CL-4B chromatography (20). A Pharmacia column (2.6 × 60 cm, bed volume 300 mL) containing Sepharose CL-4B agarose was used. Buffer containing 0.01 M phosphate buffer, pH 7.2, was used for column equilibration and elution; 2 mL incubation mixture was applied on the column. The flow rate was maintained at 20 mL/hour. The eluate was collected at 100 drops/3.4 mL/tube. The fluorescence intensity of the fractions was measured following the procedure previously described. The fractions which had higher, middle, and lower fluorescence intensities were pooled separately. These pooled fractions were concentrated by dialysis against saturated ammonium sulfate solution at 4°C overnight. The precipitate of each pooled fraction was redissolved in 0.01 M phosphate buffer (pH 7.2) and then dialyzed against 0.01 M phosphate buffer (pH 7.2) before column application.

Determination of Protein Concentration

The protein concentration was determined by a BCA protein assay on microplate following the instructions provided by the manufacturer (20).

RESULTS

Change of Fluorescence, Fructosamine, and Glycated Albumin Over Time

Because anti-AGEs are not available, AGE-associated fluorescence (excitation wavelength at 370 nm and emission wavelength at 440 nm) was used to identify AGEs (4). Since the control (BSA incubated in the absence of glucose) was incubated at the same time under identical conditions as the sample, any difference observed between the sample and the control was also used as an indication of the AGE formation.

As shown in Figure 1, when the excitation and emission fluorescence spectra of the incubation mixture (BSA + glucose) was measured one could detect an increase of the fluorescence intensity over time. Practically no AGE-associated fluorescence was observed for the BSA solution incubated in the absence of glucose. However, in the presence of glucose fluorescence was detectable after only a few days of incuba-

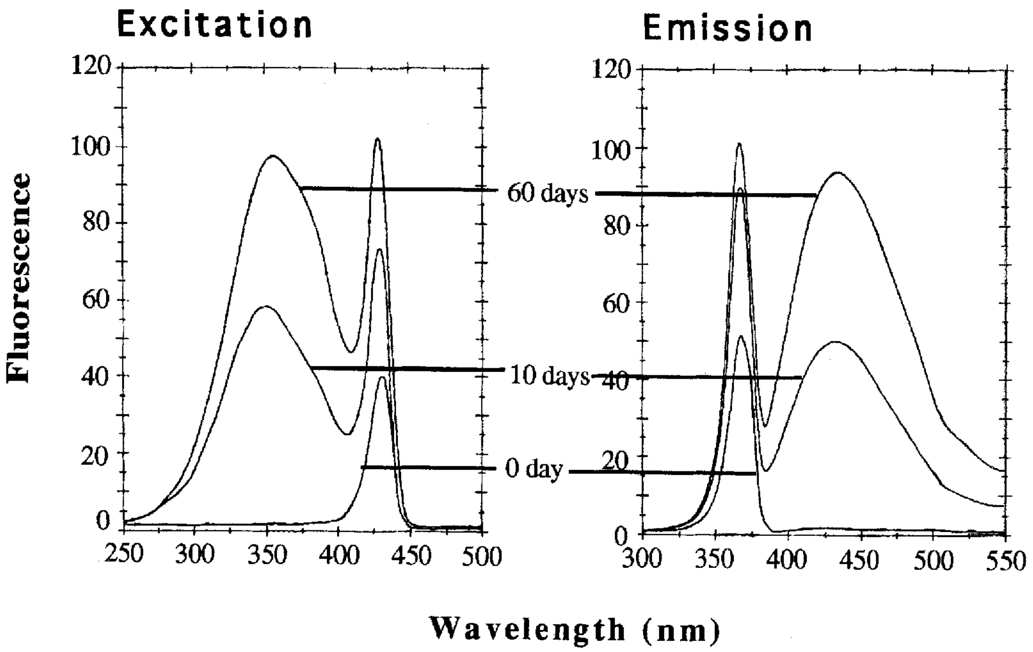

Fig. 1. Change of the AGEs-associated fluorescence spectrum during the incubation of BSA with glucose. AGEs-associated fluorescence spectrum with excitation at 370 nm and emission at 440 nm could be detected when BSA was incubated with glucose. The intensity of AGEs-associated fluorescence spectrum increased over time during incubation. No AGEs-associated fluorescence was detectable in the control.

tion indicating that the AGE-BSA aggregate apparently formed much earlier than we expected. It appears that as soon as Amadori product is formed, AGEs become detectable.

We also monitored the formation of glycated BSA (Amadori product) and fructosamine over the same time course during incubation. These results are shown in Figure 2. It is clear from the plots in Figure 2 that formation of AGEs in the incubation mixture truly started very early (Fig. 2B). Based on the measurement of AGE-associated fluorescence the rate of AGEs formation almost paralleled the formation of glycated BSA. On the other hand, no Amadori product or AGEs were formed in the absence of glucose (dotted lines).

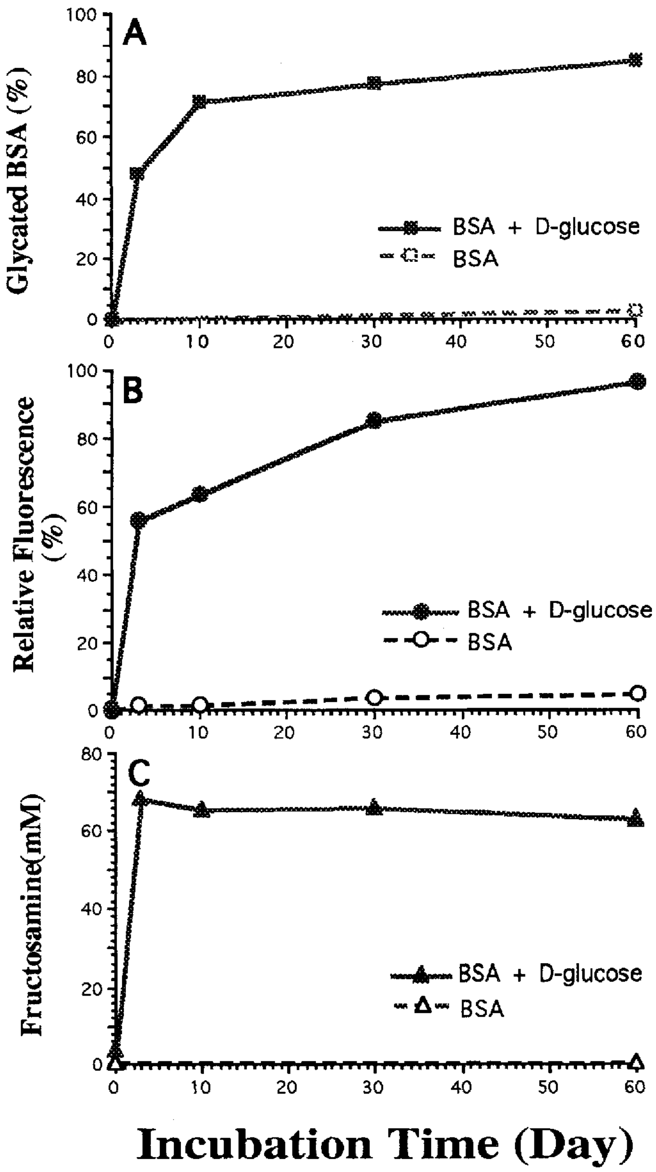

Fig. 2. The reaction between BSA and glucose was monitored by boronated gel affinity column chromatography, AGEs-associated fluorescence and fructosamine assay. The control was also monitored at the same time.

It is well known that the test for fructosamine, taking advantage of the reducing property of the Amadori product (17), essentially measures glycated BSA in serum. However, these two curves shown in Figure 2A, C do not exactly coincide with each other. Apparently, this discrepancy has something to do with the methods used for their measurements. The fructosamine assay reacts with glycated proteins but not AGEs. Conversely, the boronated gel affinity chromatography used to quantify glycated BSA in the present investigation may also react with AGEs. Exactly how this will impact the glycated BSA determination is not clear. The slow rise of the percent of glycated albumin after 10 days of incubation most likely reflects the formation of AGEs (Fig. 2A) because the formation of Amadori product (glycated BSA) was completed in three to four days incubation as shown in the fructosamine determination (Fig. 2C). It is also unclear why the decline of fructosamine was so slow considering how rapidly glycated proteins continued to form AGEs during further incubation beyond the first 3 to 4 days.

Boronated Gel Affinity Chromatography

Boronated gel affinity column is routinely used in clinical laboratories to quantify the concentration of glycated hemoglobin or albumin for monitoring glycemic control in diabetics (21). During affinity chromatography glycated proteins bind to the affinity column while the nonglycated portion passes through freely. Because we are interested in knowing whether or not glycated proteins can be separated from the AGE product by affinity chromatography we applied an aliquot from the incubation mixture at 30 days of incubation and collected the eluates to determine whether AGEs will bind to the affinity column. The AGEs were identified by the AGE-associated fluorescence measurement. We packed a larger column than that routinely used with GLYCO°GEL™ in house in order to collect sufficient protein for fluorescence measurement. As shown in Figure 3A, after 30 days incubation, only 15–20% of the total albumin remained nonglycated (peak eluted by the washing buffer). The majority of the BSA in the incubation mixture was retained by the column as glycated albumin. However, we found an additional protein peak retained by the column after eluting the glycated albumin following the regular procedure. This additional protein peak, apparently containing protein of higher affinity than the Amadori product, could only be eluted during column regeneration. Because fluorescence appears to be associated most closely with this additional protein peak, it is conceivable that the protein eluted by the regeneration buffer is the AGE, which is absent in the control. Apparently AGEs will bind to the affinity column as did its precursor, the Amadori product, but with an even higher affinity. It appears from the elution profile shown in Figure 3A that the second peak (containing the glycated BSA) also produced AGE-associated fluorescence. It is possible that some of the early product of AGEs bind to the gel with equal affinity as the Amadori product.

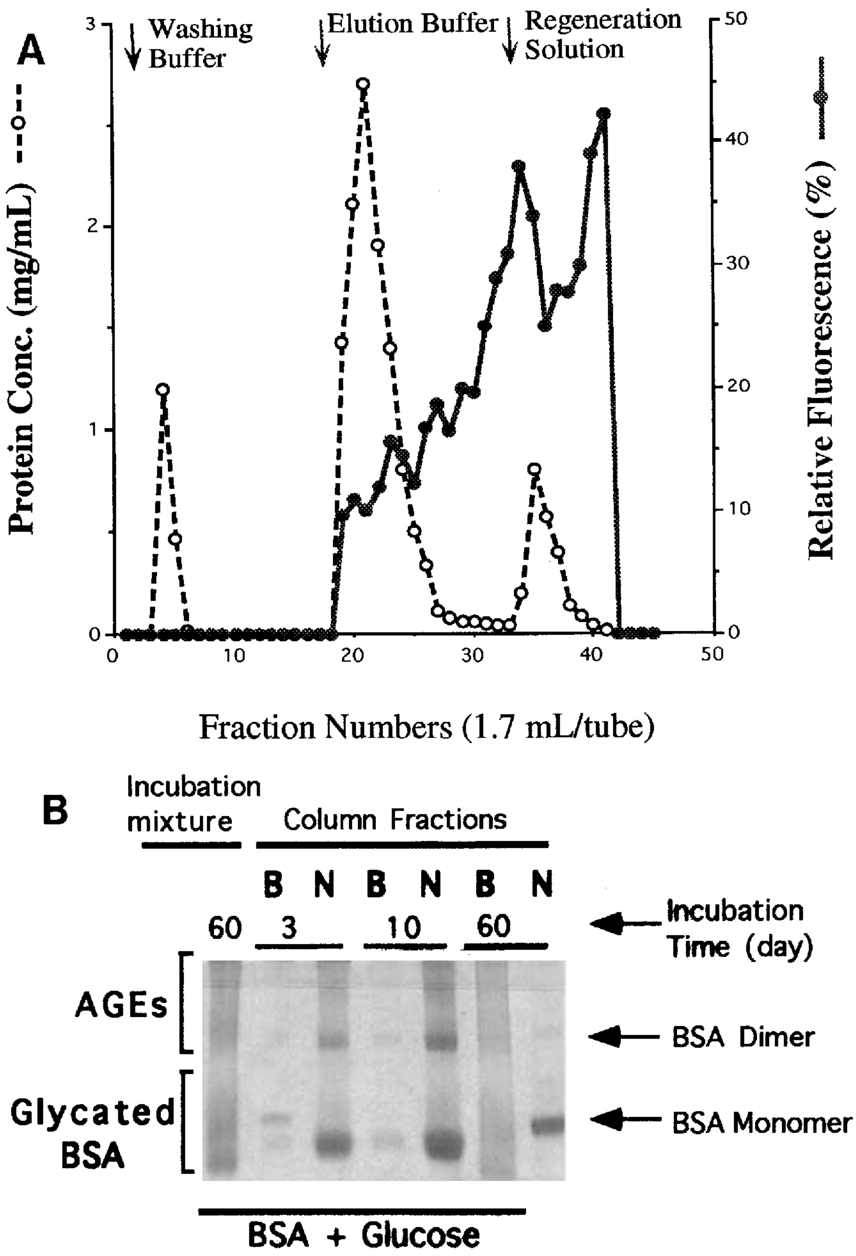

Fig. 3. Boronated gel affinity chromatography of a 30 day incubation mixture containing BSA and glucose. **A**: Elution profile of the boronated gel affinity chromatography. The first protein peak (nonbound fraction), eluted by the washing buffer represents the nonglycated portion of the BSA. The second protein peak (bound fraction) is the glycated portion of the BSA molecule. The 3rd protein peak, also associated with the highest intensity of AGEs-associated fluorescence, should be the AGE-albumin which could only be eluted by the column regeneration solution. Regarding the control, all protein was found in the first peak (not shown). **B**: Column eluates from both the bound (**B**) and the unbound fraction (**N**) were concentrated and subjected to 10% PAGEs. It appears from the PAGE pattern that the longer the incubation continued, more protein (both Amadori product and AGEs) appeared in the bound fractions and less in the unbound fractions.

Eluates from both nonbound and bound portions of the affinity column were again examined by 10% PAGE (Fig. 3B). Apparently both glycated albumin and AGEs were retained by the column (B). At the beginning of the incubation (3 and 10 days) only small amounts of proteins showed up in the bound fractions (B). At the 60th day of incubation, most BSA, both glycated albumin and AGE products, were in the bound fraction. The reverse was true for the nonbound fractions (N). The PAGE pattern of the original mixture after 60 days of incubation shows the protein bands smearing over the entire track suggesting that both the Amadori product and the AGE product are heterogeneous in size and charge.

PAGE

We also monitored changes of BSA during incubation with glucose by the PAGE technique. The electrophoretic patterns are shown in Figure 4. It appears that anywhere from 6 to 10% gel concentrations are suitable for the detection and identification of both Amadori product and AGEs of serum albumin. By comparison with the control, two major changes in the albumin can clearly be demonstrated by this technique. The first noticeable change with the albumin monomer is its electrophoretic mobility. From as early as day one of incubation, the band corresponding to the albumin monomer started to move further to the anode (indicating a higher electrophoretic mobility). The band continued to move further down-

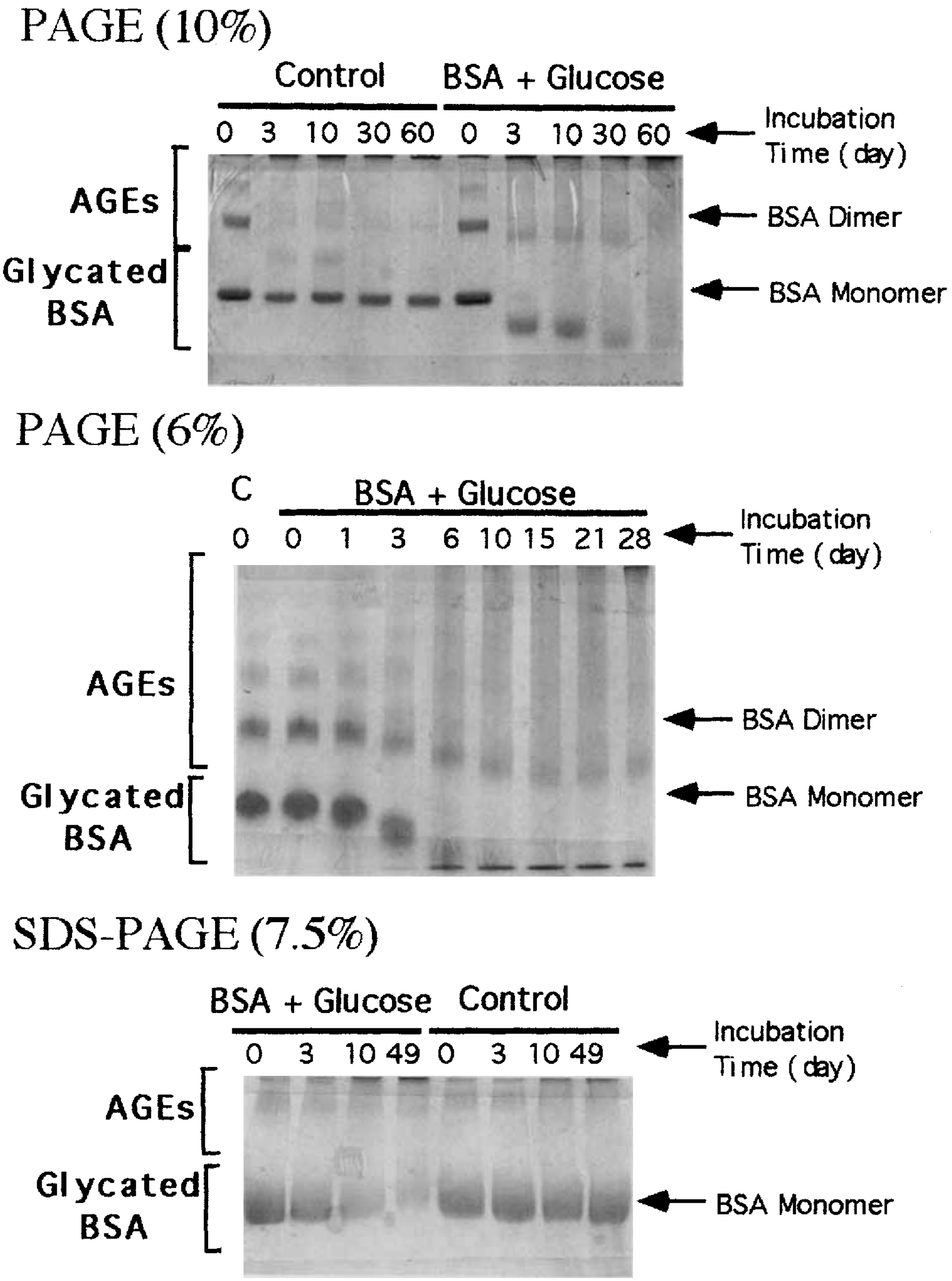

Fig. 4. Characterization of the glycation products of BSA during the incubation of BSA with glucose by PAGE in the absence and presence of SDS. Aliquots were withdrawn from the incubation mixture at various time intervals during incubation and kept frozen at –20°C. All aliquots (all contained 25 μg of protein) were thawed at 4°C at the same time and analyzed simultaneously by 10% and 6% PAGEs. They were also treated with SDS and run in the presence of mercaptoethanol to determine why the band corresponding to the albumin monomer gained higher electrophoretic mobility.

ward as incubation continued. Notice that the band not only gained higher electrophoretic mobility over time, but also became more diffuse and lighter in color intensity. At the same time another diffuse band appeared at the top of the gel suggesting the formation of cross-linking products of larger molecules. After prolonged incubation for approximately 20 to 30 days, the entire band pattern became more and more faint even though exactly 25 mg of protein was applied to every trough throughout the entire gel. When the gel was examined closely one could also observe an increasing thickness of the protein band retained at the top of the gel. Apparently, increasing amounts of larger albumin polymers were formed during prolonged incubation, and the size of these larger polymers prevented them from entering the gel. It should be noted that all of these changes observed with the mixture of BSA and glucose did not take place in the control. In the absence of glucose the monomer albumin remained practically the same in its electrophoretic mobility and intensity during the same period of time (Fig. 4, 10% PAGEs pattern). Although a small amount of BSA polymer was found on top of the gel after prolonged incubation, both the pattern and the band intensity were different from that exhibited by BSA incubated in the presence of glucose. The differences were most noticeable in the 6% PAGE pattern.

SDS-PAGE

There are two possible ways for monomer albumin to gain higher electrophoretic mobility during its incubation with glucose. One possibility involves the degradation of monomer albumin into smaller molecules; the second involves the addition of negative charges to the monomer albumin during its reaction with glucose. To differentiate between the degradation and charge increase, aliquots from the incubation mixture were subjected to SDS-PAGE. SDS-PAGE performed in the presence of a sulfhydryl agent will determine whether the BSA molecule has been degraded or not. If the size of the glycated albumin molecules remains the same as the BSA incubated in the absence of glucose, this would suggest that additional negative charges were gained by the monomer BSA during its incubation with glucose. This should account for the increased electrophoretic mobility of the molecule. As shown in the pattern of SDS-PAGEs (7.5%) in Figure 4 the monomer albumin from the mixture, incubated with glucose for various times, all moved to the same position after electrophoresis. These results indicate that the size of all monomer albumin remained the same regardless of its duration of incubation with glucose. Therefore, it is most likely that the monomer albumin became more negatively charged after it was glycated.

In the SDS-PAGE pattern we can also observe proteins on top of the gel. For the incubation mixture containing BSA and glucose the increase of the band intensity on top of the gel coincided with the gradual disappearance of the protein band derived from the monomer albumin. Notably, all the monomer albumin maintained a similar intensity in the control throughout the entire incubation period. This provides additional evidence which suggest that AGEs were formed from the glycated albumins during incubation. Apparently, AGEs are stable during treatment with SDS under reducing condition.

Size Exclusion-HPLC

Since the formation of AGEs coincides with an increase in size of the glycated protein molecules we decided to monitor the change in size of the BSA molecule by HPLC during incubation. The elution profiles of the UV tracing for both controls (BSA) and samples (BSA + glucose) are shown in Figure 5. At the 3rd day of incubation there was no significant difference in the elution profiles between the sample and the control. The major peak corresponded to the monomer BSA and a small bump at the left side of the major peak was derived from the small amount of albumin dimer. Formation of AGEs or aggregates of albumin became much more apparent at day 30. Even though there were more large aggregates with the sample there was little difference in the elution profiles between the control and the sample. It was not until the 45th of day incubation that the glycated sample started to reveal a distinctly different elution profile from that of the control. It is apparent from the UV tracings of the day 45 and day 60 incubation mixtures that more aggregates were found in the sample compared to that of the control. The broad peak of the larger aggregate suggests they are very heterogeneous in size. Small peaks also showed up after 45 days of incubation on the right side of the major peak. Since they were not observed in the control, they should represent molecules smaller than the monomer albumin, possibly the degradation products of AGEs.

The formation of aggregates much larger in size during incubation was also confirmed by PAGE patterns shown in Figure 5B. Eluates were pooled according to the four peaks designated in Figure 5A and concentrated by lyophilization and further examined by 10% PAGE. Proteins derived from peak 1 apparently are very large. They could not enter the gel because of their size and remained on the top of the gel. The majority of the proteins derived from peak 2 at day 3 of incubation were related to the dimer of the albumin but were converted to smaller AGEs at day 60 of incubation. Proteins associated with the major peak (peak 3) even at day 3 gained higher electrophoretic mobility, moving ahead of the original monomer albumin during electrophoresis without changing in size (as shown in the size exclusion-HPLC elution profile). The pattern from the day 3 mixture not only confirms our fructosamine measurement suggesting that the majority of the BSA was glycated at day 3 but also supports our SDS-PAGE study implying that the size of the BSA molecule remained the same after 3rd day incubation. The results also

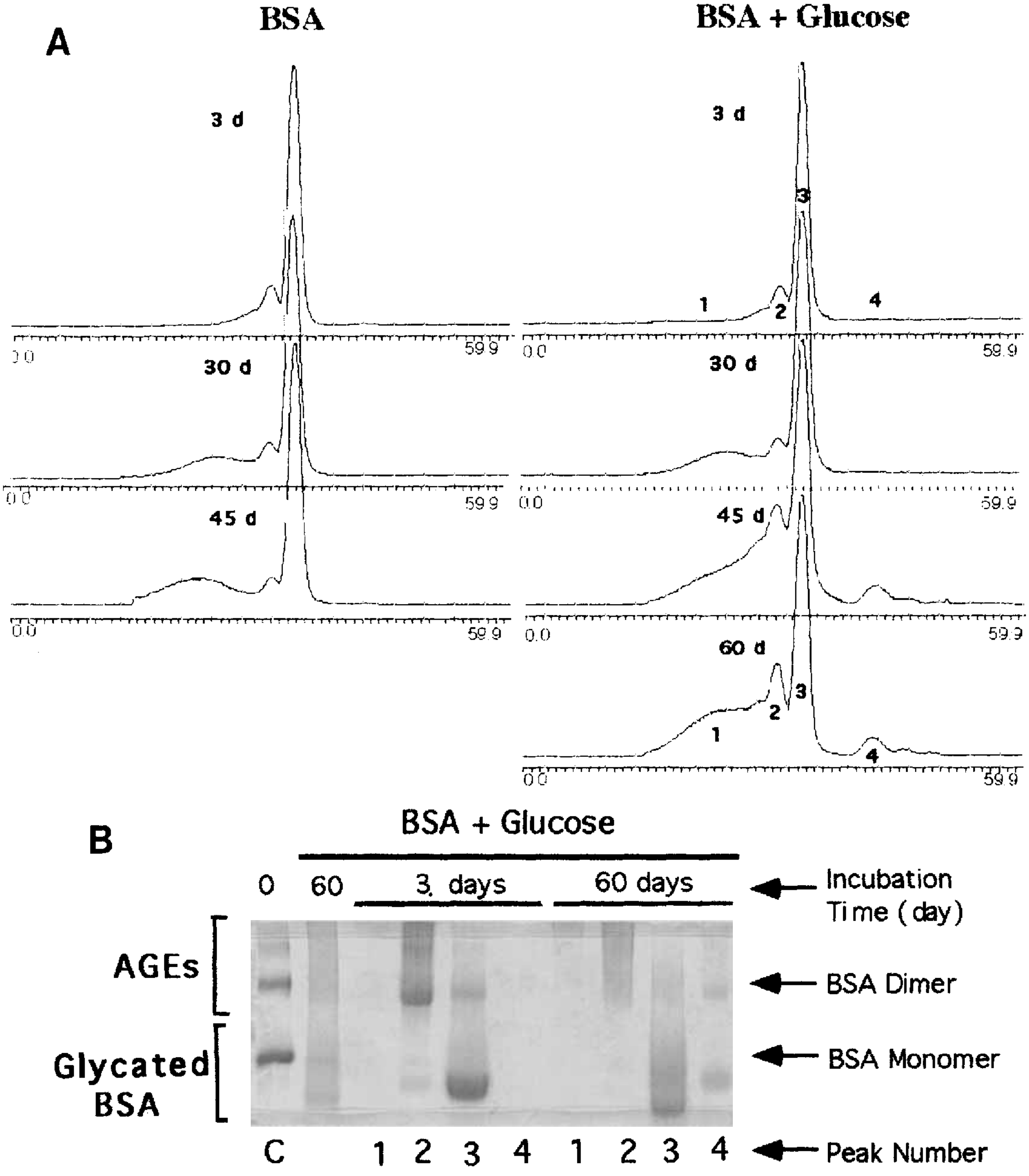

Fig. 5. Change of the size of monomeric BSA molecule during incubation of BSA with glucose monitored by size-exclusion HPLC. Aliquots withdrawn from the incubation mixtures at various time intervals were applied to the HR 6R column on HPLC. **A**: The UV absorbance of the eluates were traced continuously during HPLC. Notice that formation of polymers was more rapid and extensive in BSA incubated in the presence of glucose than in its absence. Polymers of BSA of varying size beyond the dimmer were grouped under peak 1. **B**: Eluates corresponding to various peaks were collected and were subjected to 10% PAGEs.

Incubation (d)	Peak number	Total protein applied (μg)	Incubation (d)	Peak number	Total protein applied (μg)
3	1	2	60	1	4
3	2	24	60	2	11
3	3	25	60	3	25
3	4	9	60	4	10

reinforce the idea that the higher electrophoretic mobility of the glycated proteins was most likely associated with the increased negative charge derived from glycation. At day 60, proteins associated with the peak 3 become more heterogeneous in electrophoretic mobility. The size and the nature of proteins associated with peak 4 remain unclear to us. We could not make any sense out of the PAGE pattern for protein at peak 4 at 60 days of incubation: for these molecules to display such a PAGE pattern, the molecule of peak 4 would have to carry fewer negative charge than monomer albumin.

Sepharose CL-4B Chromatography

In order to relate the AGE-associated fluorescence with the size of BSA during incubation we performed Sepharose CL-4B chromatography to allow the application of a larger aliquot of the incubation mixture in order to collect sufficient amounts of protein from the column for fluorescence measurement. Because Sepharose CL-4B chromatography separates globular proteins in the molecular weight range between 6×10^4 and 2×10^7 daltons, it is especially suited to study BSA polymerization.

As shown in the elution profile in Figure 6 a fluorescence peak was detected as early as day 3 of incubation even though there was no obvious change of the major protein peak. Apparently the small amount of AGEs formed at day 3 was too little to be seen as a separate protein peak though it could be detected by the AGE-associated fluorescence. The finding shown in Figure 6A agrees with results shown in Figures 2, 3B and 5B, which all suggest that formation of AGE-albumin occurs very early during incubation with glucose. At day 50 of incubation intense fluorescence was detected and the fluorescence peak appeared much earlier during elution. The data

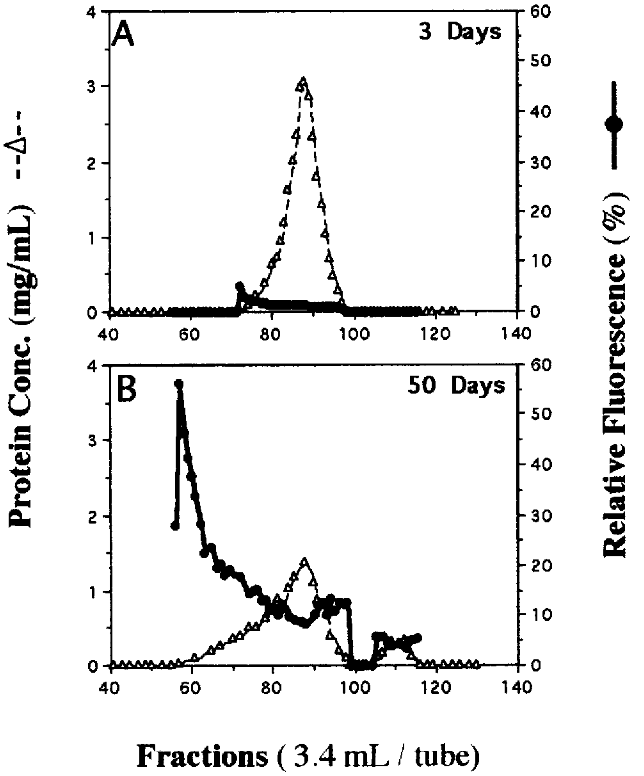

Fig. 6. Sepharose CL-4B column chromatography of a mixture containing BSA and glucose after incubation for 3 and 50 days. **A**: The intensity of the AGE-associated fluorescence was correlated with the size of the albumin molecules separated by the Sepharose CL-4B chromatography.

seems to suggest that not only the amount but also the size of AGE-albumin was increased upon further incubation. The position of the fluorescence peak suggests that the molecular weight of the largest AGE polymer could exceed millions. Since the molecular weight of monomer albumin is approximately 70,000 daltons, the largest AGE polymer could consist of multiple numbers of albumin molecules. Notice that the peak corresponding to the monomer albumin also became smaller and skewed toward higher molecular weight. As we described in Figure 5A there was a small peak corresponding to a smaller molecule than the original monomer albumin that started to appear at 50 days incubation. The nature of these proteins remains unclear to us.

Anion Exchange Chromatography

We know that upon glycation, the electrophoretic mobility of the BSA molecule increases because the molecule becomes more negative charged. However, it is not clear whether AGE-BSAs will also change their charge property. To investigate, we applied aliquots from the incubation mixture at various time intervals to anion exchange column on HPLC. The elution profiles are shown in Figure 7. It is clear from these elution profiles that new proteins carrying more negative charges begin to appear during incubation (Fig. 7B, C). It should be noted that more negatively charged molecules are eluted later by buffers containing higher salt concentrations. As shown in Figure 7 as the incubation proceeded the charge property of these new negatively charged molecules also continued to change. Notice that none of these new protein peaks were observed in the control (Fig. 7D).

Again in order to relate AGE-associated fluorescence with proteins separated by the anion exchange chromatography, a larger DEAE Sephacel column was packed to collect sufficient proteins for fluorescence measurement. It is clear from the elution profile in Figure 8 that AGE-associated fluorescence was associated with proteins carrying more negative charge. It also appears that the more negatively charged molecules were associated with higher fluorescence intensity.

Chromatofocusing-HPLC

To confirm the results obtained from anion exchange chromatography, aliquots from day 50 incubation mixtures were also examined by chromatofocusing. The AGE-associated fluorescence was also measured to relate various protein molecules in the eluates. The elution profile in Figure 9 confirms that AGEs formed during incubation are heterogeneous in charge. Based on the protein determination, proteins with pI values different from monomeric BSA were formed during incubation. Some BSA molecules apparently became so much more acidic after glycation that the pI of the glycated albumin dropped to less than pH 4; they could only be eluted by 1 M NaCl aqueous solution at the end of pH 4 gradient. The results of fluorescence measurement also suggest that

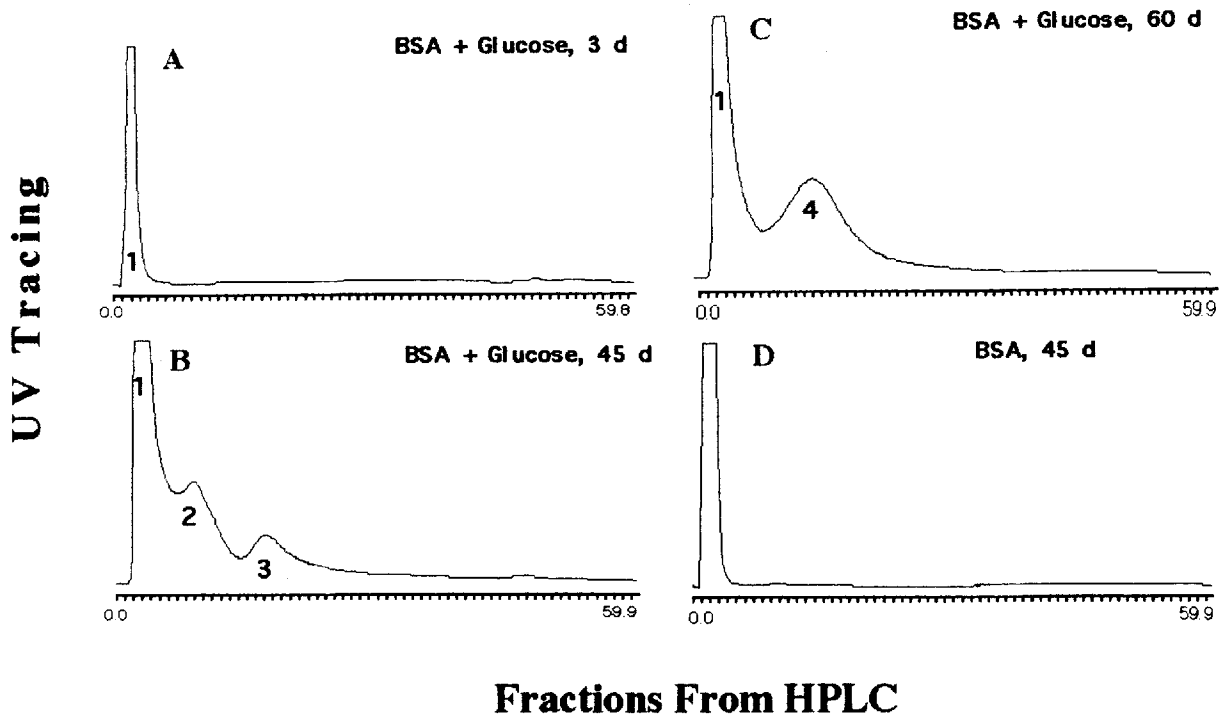

Fig. 7. Change of the charge property of BSA during its incubation with glucose was monitored by anion exchange-HPLC. **A**: These are the UV tracings recorded continuously during HPLC. New peaks appeared at the right-hand side of the original albumin peak (BSA monomer) and were derived from albumin molecules after becoming more negatively charged during incubation in the presence of glucose. No new peaks were observed in the control. Notice that the size of these new protein peaks also increased over time during incubation. **B**: Aliquots from the control (0d, 25 μg; 60 d, 25 μg) and the sample at 3 days (peak 1, 1.8 μg; peak 2, 23 μg; peak 3, 26 μg; peak 4, 0.7 μg) and 60 days (peak 1, 9 μg; peak 2, 11 μg; peak 3, 25 μg; peak 4, 3.6 μg) were further analyzed by 10% PAGE.

each protein peak may comprise more than one type of molecule with different fluorescence intensity. However, from the elution profile it appears that a large portion of the BSA molecules had the same pI (pH 4.7) as the original monomeric albumin. The fluorescence measurement suggests that BSA with pI at 4.7 is a mixture of heterogeneous molecules. Some of these molecules could be AGE-albumin.

DISCUSSION

It appears that AGEs can be detected in the first few days in a mixture containing BSA and an excess amount of glucose as soon as the Amadori product or glycated BSA is available. The initial reaction of glucose with BSA apparently alters the charge property of the protein in such a way that the pI of the glycated protein will change to a lower pH and the protein will migrate faster in an electric field. The migration pattern on the polyacrylamide gel indicates that the monomeric BSA in the incubation mixture continued to increase its negative charge during incubation. At the same time a portion of the glycated BSA was converted to AGEs. Because an albumin molecule contains multiple arginine and lysine residues, it is conceivable that more than one free amino group are available to react with glucose. It also means that multiple sites on the albumin molecule are available for reacting with glucose and multiple sites are available for cross-linking with other protein(s). During the incubation with glucose varying degrees of glycation may occur to individual albumin molecules resulting in varying degrees of cross-linking and also leading to molecules heterogeneous in charge and size. Whether this charge heterogeneity in glycated albumin would also have an impact on the rate of AGE formation is not known. AGEs are not stationary: once formed they continue to cross-link with other proteins and undergo additional rearrangement. Consequently, the size and charge of AGEs also continue to change. In both the 10% and 6% PAGE some of the largest AGE molecules could not enter the gel and remained on top of the sample trough. Some AGEs did enter the gel appearing as smeared, diffused bands. Therefore, our analyses clearly suggest that AGE molecules are heterogeneous in size and charge properties and are different in AGE-associated fluorescence intensity. When the fluorescence is expressed per mg protein, the AGEs with the largest molecular weight were always associated with highest fluorescence intensities. It appears that when AGEs continue to grow in size that new linkages between the glucose and the protein moiety are the major contributors in increased fluorescence intensity.

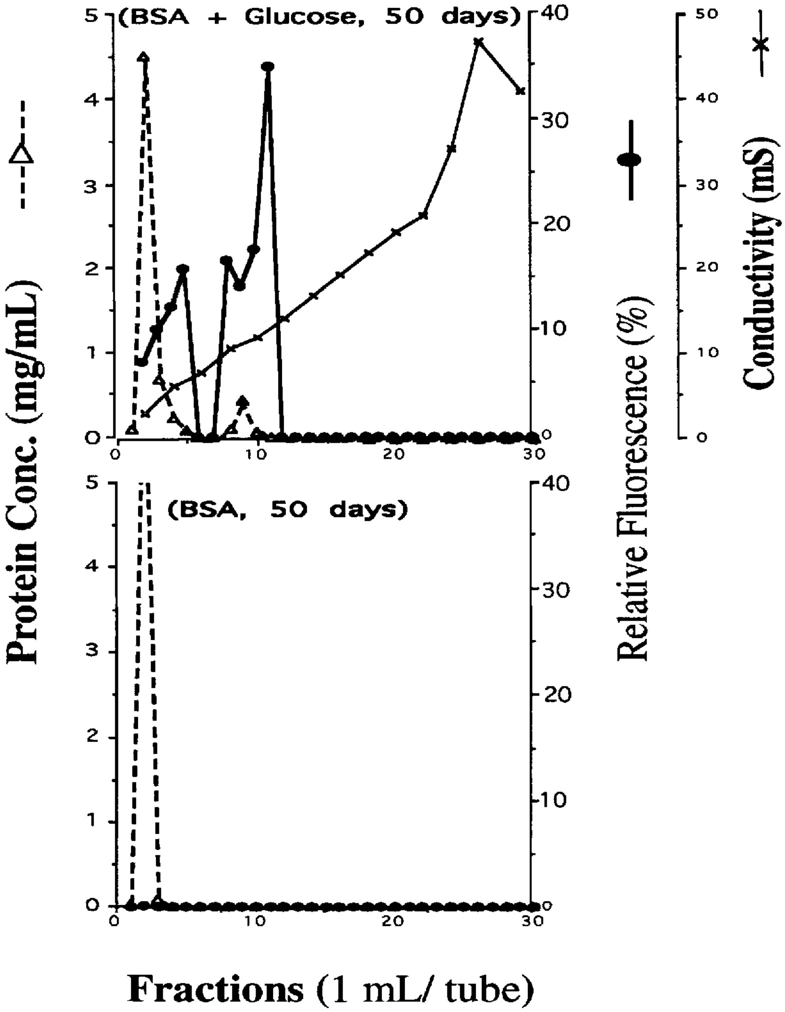

Fig. 8. DEAE chromatography of albumin incubated in the presence of glucose for 50 days. The AGEs-associated fluorescence was measured to correlate with the albumins of different negative charges separated by the DEAE chromatography. No AGE-associated fluorescence and more negatively charged albumin were detected with the control.

AGE is a potentially useful marker for monitoring glycemic control, predicting the risk of diabetic and aging associated clinical complications, and monitoring the treatment of patients with micro- and macro-vascular diseases, including retinopathy, atherosclerosis, nephropathy and neuropathy (3). Once glycated proteins are formed, improvement of glycemic control alone will not be sufficient to prevent the continued progression of pathologic processes such as the formation of AGEs from glycated proteins and subsequent tissue damage. In this case, pharmacologic agents, such as aminoguanidine, should be used in conjunction with glycemic control to interfere directly with the self-perpetuating process of AGE-formation and the accumulation of additional glycated proteins.

Conceivably, an assay for AGEs would be useful for monitoring the treatment of patients with diabetes and aging related clinical complications. Assays for AGEs should also be valuable for assessing the efficacy of new drugs (8). Measurement of tissue AGE-protein has been recommended to replace the less accurate, technically difficult measurement of the thickness of capillary basement membrane in determining the severity of vascular diseases and assessing the risk for various clinical complications (3, 9). However, no commercial kit is available at this time.

Apparently isolation and purification of AGE is the first step for the establishment of an immunoassay for AGEs. Because AGEs differ so dramatically in size and charge from their precursors, isolation of AGE is not a problem. Both ion exchange and gel filtration chromatographies are useful for this purpose. However, it is a challenge to select correctly among all the heterogeneous AGE molecules for the preparation of anti-AGE antibodies or for use as control or calibrator for the assay. In the present study we incubated the BSA with glucose for up to 60 days. Based on our PAGE analyses, there were still some Amadori products present in the incubation

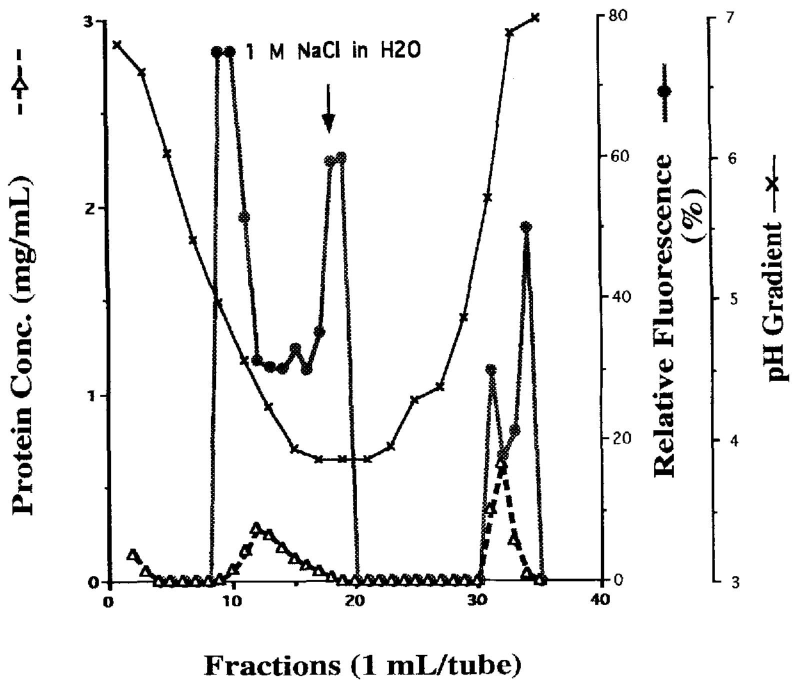

Fig. 9. Characterization of a 50 day incubation mixture by chromatofocusing. **A**: One hundred microliters of treated sample were injected onto the column elution profile of a pH gradient from pH 7 to pH 4. At the end of the regular elution at pH 4 1 M NaCl aqueous solution was used for column regeneration. Any protein with pI less than pH 4 will be eluted by the 1 M NaCl solution. **B**: Aliquots from the control mixture (C, 25 µg) and the sample, both at 60 days of incubation, plus eluates collected from incubation mixtures at 3 (protein peaks eluted at pH 4.3, 25 µg; and by 1 M NaCl, 5 µg) and 50 days of incubation (protein peak by 1 M NaCl, 21 µg), after chromatofocusing were further analyzed by 10% PAGEs. Both protein peaks eluted at pH lower than original albumin. Both peaks are shown to contain both glycated proteins and AGEs.

mixture not converted to AGE; and the AGE molecules continued undergoing changes in size and charge. Whether it is necessary to identify the physiological active AGEs based on their reaction with the AGE specific receptor from endothelial cells and monocytes (13, 22) is something we need to consider later. A different approach would be to select those AGEs having the same characteristics as the AGEs appearing in blood circulation or in urine from patients with diabetes.

It is surprising to find that after undergoing all the molecular rearrangements from glycated protein to AGE that AGE still binds to the boronated affinity column and with even a higher affinity. The continuous conversion of glycated protein to AGE and the binding of AGE with boronated gel affinity column (AGEs are not eluted by regular eluting buffer) may result in a slight underestimation of the percent of glycated protein in the blood. It would require a careful study to know realistically how such phenomena affect the estimation of, for example, percent of glycated hemoglobin routinely used for monitoring glycemic control. Because the fructosamine assay does not react with AGE its result may also be impacted by AGE formation.

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